

Motivation

- VLMs combine visual and language reasoning, enabling robots to generate grounded actions from natural-language instructions.
- VLMs need to adapt when the **environment changes** or when actions become uncertain.
- Many existing frameworks react only after a failure (e.g., due to **latency**), which can lead to unnecessary movements or collisions.

ProAct-VLM

- Adaptive closed-loop framework for task planning.
- VLMs operate with continuous perception feedback.
- Interrupts affected actions and triggers re-planning.

Experimental Results Across VLMs

Disturbance Scenarios



Scenario	GPT-4o		Gemini 2.5		Claude 4		Qwen 2.5		Llama 4	
	Succ	Eff	Succ	Eff	Succ	Eff	Succ	Eff	Succ	Eff
S1	100	100	100	100	100	100	20	100	100	100
S2	100	100	100	100	40	100	70	100	60	97
S3	80	80	100	100	0	0	60	70	100	100
S4	80	92.5	100	100	0	0	40	80	10	50
S5	100	96.6	100	100	0	0	50	90	100	96
S6	90	95	100	100	0	0	40	75	90	100
S7	70	84	100	100	10	66	0	0	30	83
Avg.	89	93	100	100	21.4	38	40	73.6	70	90

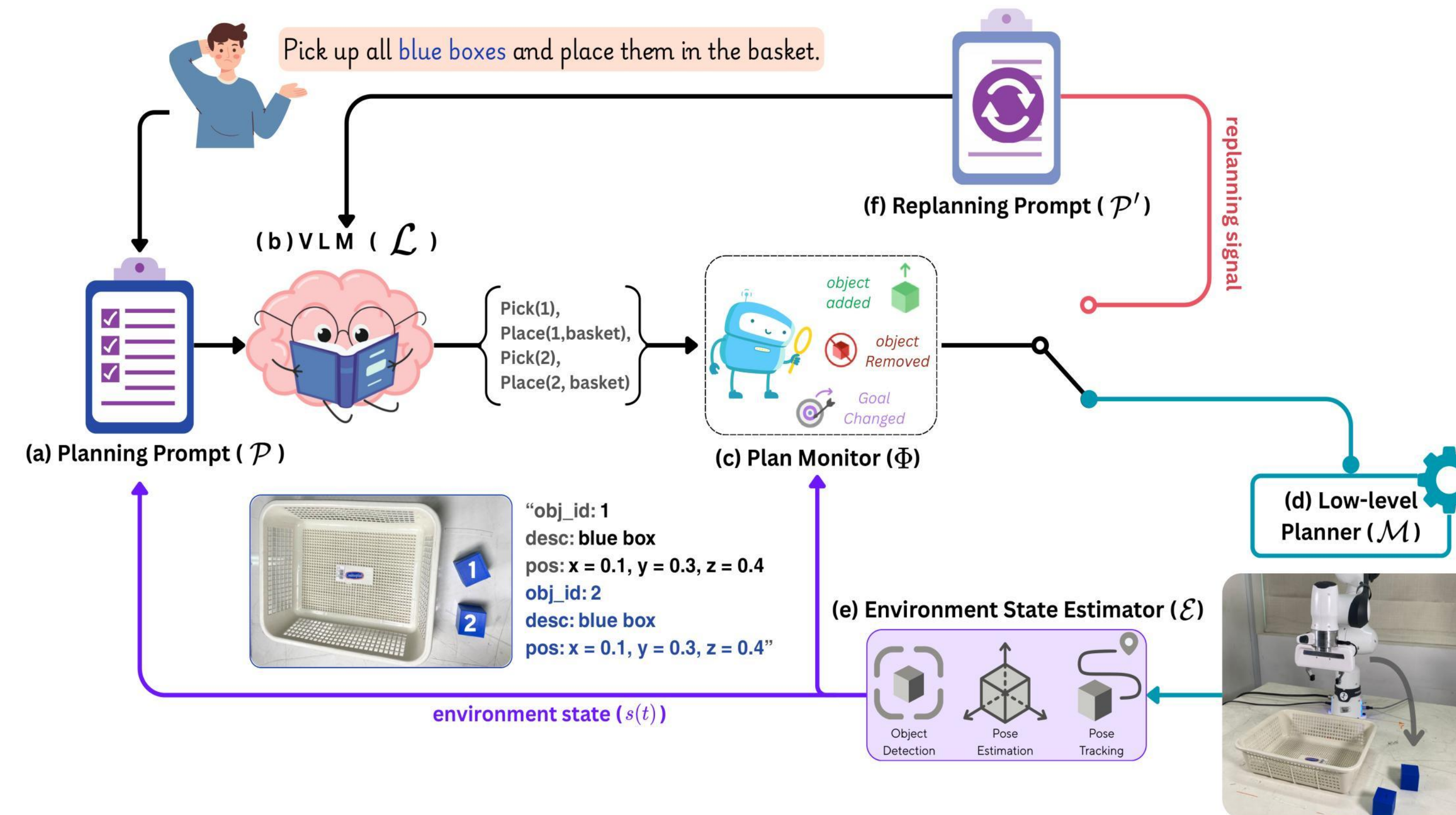
Experimental Setup

- Franka robotic arm
- YCB objects
- 3 disturbance types form 7 test scenarios

Evaluation Metrics

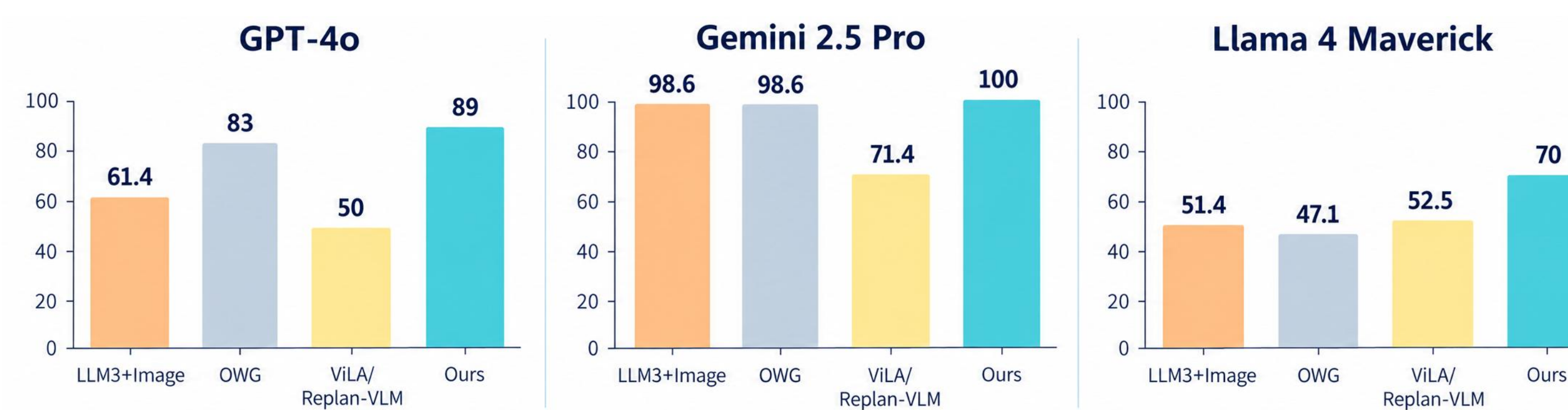
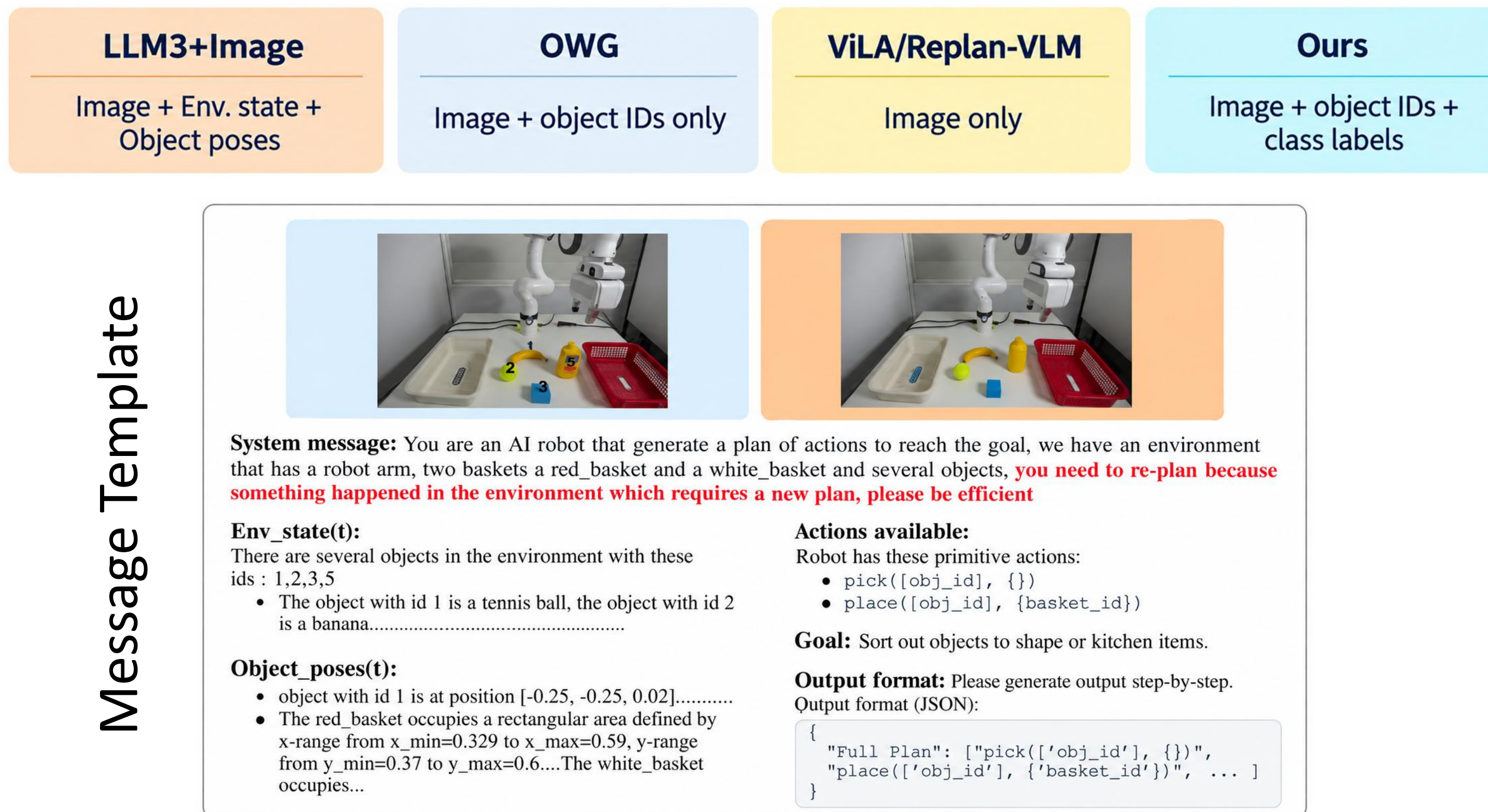
- Success rate
- Planning efficiency

ProAct-VLM

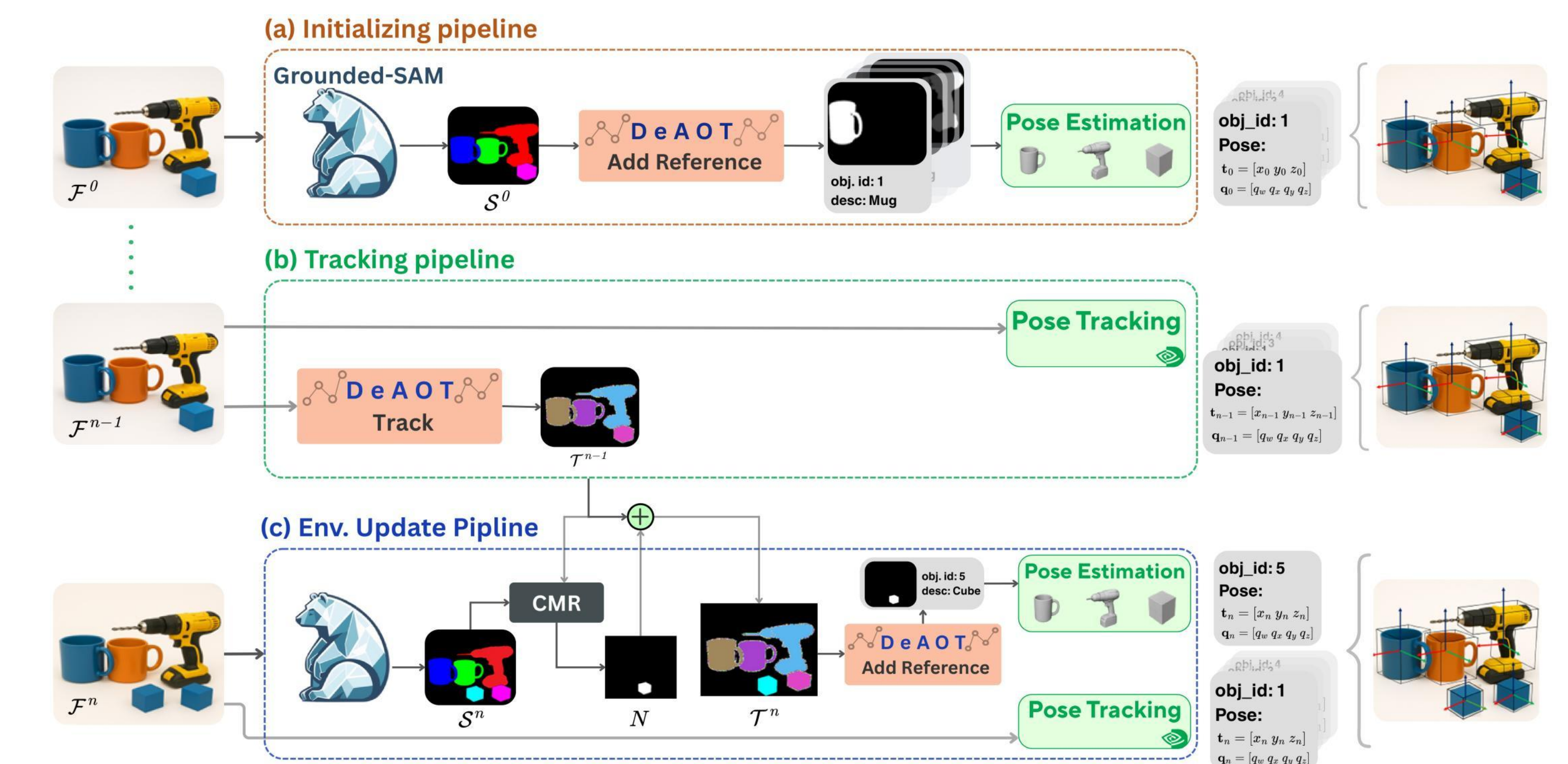


Comparative analysis (Grounding)

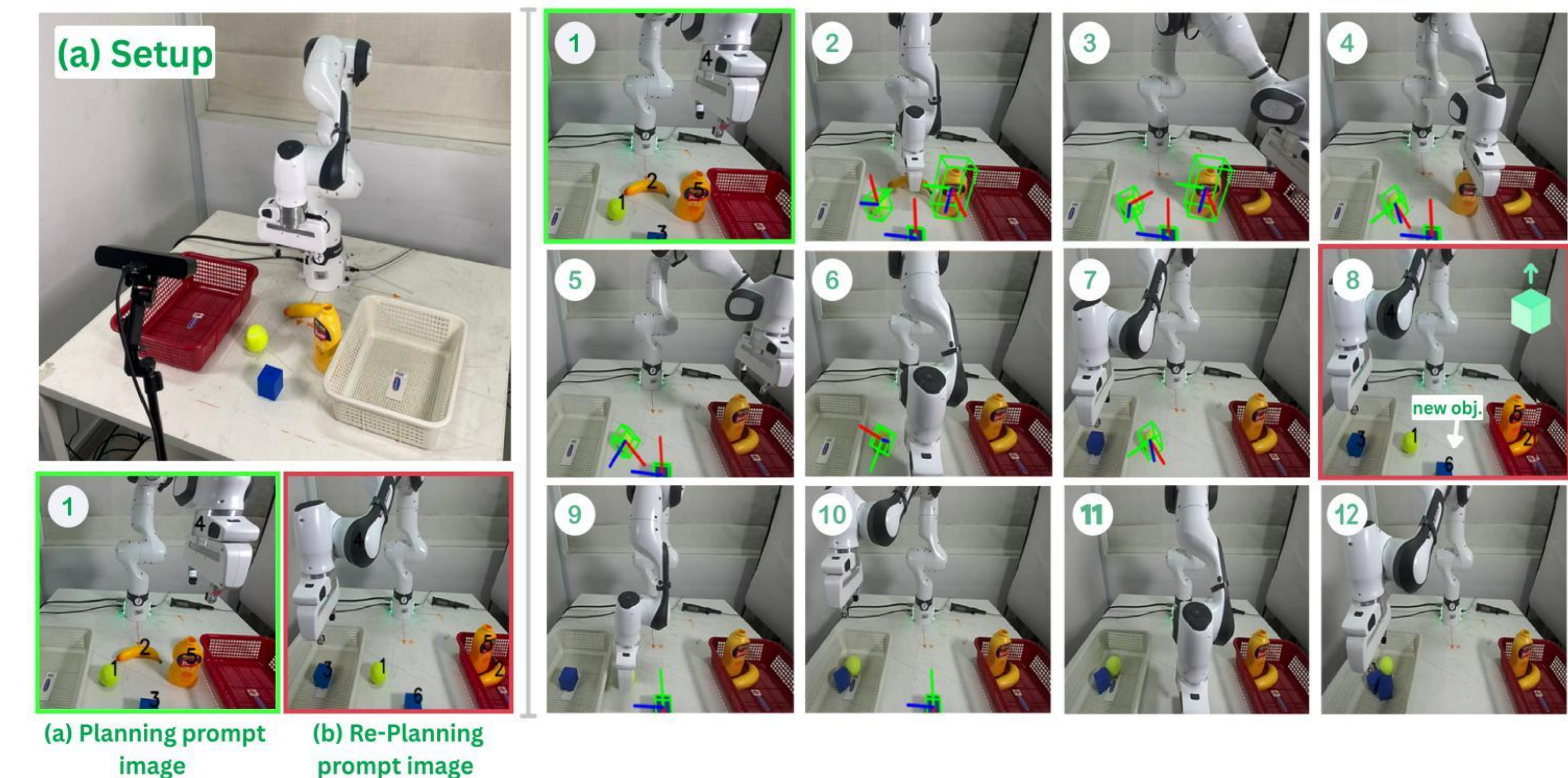
We evaluate how grounding strategy affects planning performance (1) Compare 4 grounding strategies. (2) Evaluate across 3 VLMs and 7 scenarios. (3) Fixed: system message, actions, goal.



Environment state estimator ε



Real Robot Execution



Conclusion

- Continuous perception is necessary.
- Detects changes and re-plans before failure.
- Feed fresh environment state through real-time tracking and pose estimation.

